

ProjectSync Telegram Bot Setup Guide

Sends hourly reminders, deadline warnings, and daily digests to students worldwide — in their language and timezone.

Quick Start (5 Minutes)

Step 1: Create Telegram Bot

1. Open Telegram and message [@BotFather](#)
2. Send `/newbot`
3. Name it: ProjectSyncBot (or your choice)
4. Username: projectsync_bot (must end in bot)
5. **Copy the API token** (looks like: 123456789:ABCdefGHIjkLMNOpqrsTUVwxyz)

Step 2: Configure Your Bot Token

Open `telegram_bot.py` and replace:

```
BOT_TOKEN = "YOUR_BOTFATHER_TOKEN_HERE"
```

Also replace your Supabase credentials:

```
SUPABASE_URL = "https://YOUR_PROJECT_ID.supabase.co"
SUPABASE_KEY = "YOUR_SERVICE_ROLE_KEY_HERE" # NOT the anon key!
```

⚠ Important: Use the **Service Role Key** (starts with `eyJ...`), not the Anon Key. Find it in Supabase Dashboard → Settings → API → `service_role` key.

Step 3: Install Dependencies

```
pip3 install supabase pyjwt pytz aiohttp
```

Step 4: Test Run

```
cd ~/Downloads/projectsync # or wherever your files are
python3 telegram_bot.py
```

You should see output like:

```
=====
ProjectSync Telegram Bot
Started at: 2026-07-18 13:00:00
=====
    Sent hourly reminder to user@example.com
    Sent deadline warning to student@school.edu for 'Biology Report'
=====
Finished at: 2026-07-18 13:00:05
=====
```

Step 5: Set Up Cron Job (Run Every Hour)

On macOS, you have two options:

Option A: Cron (Simple)

```
# Open crontab editor
crontab -e

# Add this line (runs every hour at :00):
0 * * * * /usr/bin/python3 /Users/YOUR_USERNAME/Downloads/projectsync/telegram_bot.py >> /Users/
↳ YOUR_USERNAME/Downloads/projectsync/bot.log 2>&1

# Save and exit (Ctrl+O, Enter, Ctrl+X if using nano)
```

Replace `/Users/YOUR_USERNAME/Downloads/projectsync/` with your actual path.

To find your Python path:

```
which python3
```

Option B: launchd (More Reliable on macOS) Create a plist file:

```
nano ~/Library/LaunchAgents/com.projectsinc.bot.plist
```

Paste this (update paths):

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>Label</key>
  <string>com.projectsinc.bot</string>
  <key>ProgramArguments</key>
  <array>
    <string>/usr/bin/python3</string>
    <string>/Users/YOUR_USERNAME/Downloads/projectsync/telegram_bot.py</string>
  </array>
  <key>StartCalendarInterval</key>
  <array>
    <dict>
      <key>Minute</key>
      <integer>0</integer>
    </dict>
  </array>
  <key>StandardOutPath</key>
  <string>/Users/YOUR_USERNAME/Downloads/projectsync/bot.log</string>
  <key>StandardErrorPath</key>
  <string>/Users/YOUR_USERNAME/Downloads/projectsync/bot_error.log</string>
</dict>
</plist>
```

Load it:

```
launchctl load ~/Library/LaunchAgents/com.projectsinc.bot.plist
launchctl start com.projectsinc.bot
```

Check if it's running:

```
launchctl list | grep projectsinc
```

Free Cloud Hosting Alternatives (If Your Mac Isn't Always On)

Service	Free Tier	How To
PythonAnywhere	1 scheduled task, always on	Upload file, set daily schedule
Railway.app	\$5 credit, 500 hours	Connect GitHub repo, auto-deploy
Render.com	750 hours/month	Create background worker from repo
GitHub Actions	2,000 min/month	Schedule workflow to run hourly

GitHub Actions (Recommended Free Option)

Create `.github/workflows/telegram-bot.yml` in your repo:

```
name: Telegram Bot
on:
  schedule:
    - cron: '0 * * * *' # Every hour
  workflow_dispatch:

jobs:
  run-bot:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-python@v4
        with:
          python-version: '3.11'
      - run: pip install supabase pyjwt pytz aiohttp
      - run: python telegram_bot.py
    env:
      BOT_TOKEN: ${ secrets.BOT_TOKEN }
      SUPABASE_URL: ${ secrets.SUPABASE_URL }
      SUPABASE_KEY: ${ secrets.SUPABASE_KEY }
```

Then in GitHub repo → **Settings** → **Secrets and variables** → **Actions**, add:

- BOT_TOKEN
- SUPABASE_URL

- SUPABASE_KEY

How Students Link Their Telegram

1. Student opens your ProjectSync app
2. Goes to **Profile** → **Link Telegram**
3. Opens @ProjectSyncBot on Telegram
4. Presses **START**
5. Bot replies with their Chat ID
6. Student pastes Chat ID into the app
7. Sets reminder hours (e.g., 4 PM - 10 PM)
8. Done! They'll receive reminders in their language

Bot Commands (Optional Enhancement)

You can add these commands in BotFather:

```
start - Start using ProjectSync bot
status - Check your pending tasks
snooze - Snooze reminders for 1 hour
settings - Change reminder hours
help - Get help
```

Troubleshooting

Problem	Solution
ModuleNotFoundError: No module named 'supabase'	Run <code>pip3 install supabase</code>
Bot not sending messages	Check <code>BOT_TOKEN</code> is correct
Wrong timezone reminders	Check user' s timezone in Supabase <code>profiles</code> table
Messages in wrong language	Check <code>language</code> field in <code>profiles</code> table
Cron not running	Check <code>bot.log</code> for errors; verify path to <code>python3</code>

What the Bot Sends

Type	When	Content
Hourly Check-in	Every hour during user' s preferred hours	List of pending tasks with due dates
Deadline Warning	When task due within 24h	Urgent alert with task name and time left
Daily Digest	7 AM user' s local time	Summary of all tasks due today
Task Assigned	When someone assigns a task	Notification with task details

Table 3 – continued

Type	When	Content
Task Completed	When teammate marks done	Team update on progress

All messages are translated to the user's selected language automatically!